WATTWISE

Keep the lights on. Keep the peak down.

WattWise watches every appliance in a restaurant and gently shifts the ones that can wait, so the electricity peak stays under a limit the owner picks. Service never stops.

Built at Cal Hacks 13.0 by Kritazya, Suyog and Ashirwad, with Fetch.ai agents, Claude, SpacetimeDB, Convex and Microsoft Azure.

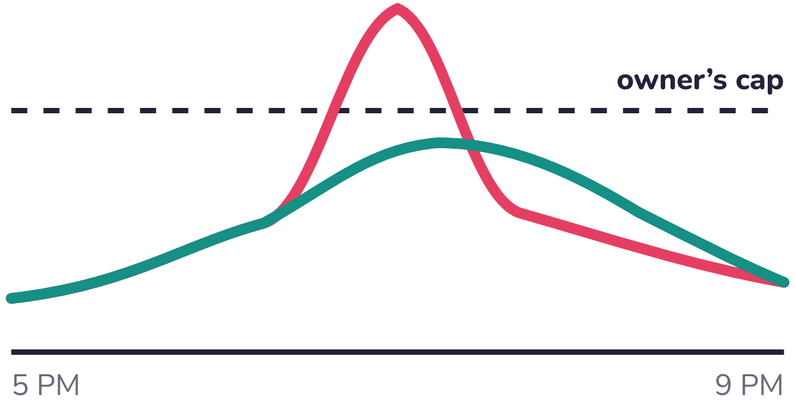
- **The fryer, grill and oven keep cooking**
- **Fridges never drop below a safe level**
- **The dishwasher waits a few minutes when it helps**

THE PROBLEM

One busy 15 minutes can set the whole month’s bill.

Many businesses pay for electricity twice. Once for the energy they use, and again for their **demand**: the highest 15 minute average power anywhere in the month.

In a restaurant that moment is usually the Friday dinner rush, when the fryer, oven, dishwasher and air conditioning all pull hard at once. Nobody sees it happen. It shows up weeks later on the bill, and that one moment sets the demand charge for the entire month.



Without control: one rush sets the bill

With WattWise: same work, spread out

Sketch, not data

BACKGROUND RESEARCH

Restaurants are hungry for power and short on margin.

5 to 7×

more energy per square foot than other commercial buildings, and up to 10× for busy quick service spots.[1]

~5 million

US commercial customers could sign up for rates with demand charges above $15 per kW, out of about 18 million.[2]

30% to 70%

of a commercial electricity bill can come from demand charges, according to reporting on NREL's survey.[3]

A few cents

on the dollar is what many restaurants keep. National Restaurant Association data put casual dining near 3% operating margin.[4]

What we noticed: a lot of the peak comes from equipment that could wait a few minutes without anyone noticing, like a dish cycle, an ice batch or a degree of cooling.

Sources on the last slide

HOW WE SOLVE IT

Sense, decide, act. Every minute.

1

Owner sets the rules

Each appliance is always on, flexible or deferrable, with limits. The owner picks a peak cap.

2

Agents bring context

Fetch.ai agents fetch weather, grid prices and nearby events, like a Giants game down the street.

3

One hub reports

Every appliance reports to one ESP32 style hub each minute. Commands go back the same way.

4

The engine plans

It forecasts the next blocks and moves flexible load: hold the dishwasher, trim the HVAC a little.

5

Claude handles the hard calls

When no move inside the rules is enough, Claude picks one of the engine’s options within the owner’s authority.

WHY OWNERS CAN TRUST IT

The owner holds the keys.

Tested code computes every number

kW, dollars and forecasts come from the engine. The AI only quotes them and never invents one.

Rules are hard limits

Always on equipment is never touched, and refrigeration never goes below 50%, not even in an emergency.

Energy is moved, not cut

Anything held back is caught up the same day. The owner can undo any decision.

It works without the AI

Switch Claude and every agent off and the kitchen runs the same, with safe defaults.

SOURCES

Background research sources

[1] ENERGY STAR for Small Business: Restaurants. energystar.gov/buildings/resources-audience/small-biz/restaurants

[2] NREL (McLaren et al., 2017), A Survey of U.S. Demand Charges. nrel.gov/docs/fy17osti/68963.pdf

[3] Energy-Storage.news on the NREL demand charge survey. energy-storage.news/news/energy-storage-could-have-significant-impact-on-us-businesses-energy-costs

[4] Chron, Operating Margins for Restaurants (National Restaurant Association data). smallbusiness.chron.com/operating-margins-restaurants-21413.html